

Enhancing Value Decomposition With Target Transformation in Cooperative Multi-Agent Reinforcement Learning

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Abstract—The increasing need for cooperation among intelligent machines has heightened the importance of cooperative multi-agent reinforcement learning (MARL). However, a dominant class of cooperative MARL approaches relies on monotonic value decomposition, which enables scalable decentralized execution but restricts the representable class of joint action-values. However, existing remedies bias learning targets toward high-value samples, which can be fragile under stochastic returns because optimistic emphasis may amplify lucky but suboptimal trajectories. To solve this challenge, we propose Target Transformation, which maps non-monotonic and stochastic learning targets into a monotonic-representable surrogate while preserving the optimal joint action. Building on this idea, we develop Uncertainty-aware Target Transformation (UT2) with value-based and policy-based instantiations that combine an uncertainty estimator with a best-individual coordination envelope. Experiments on diverse cooperative MARL benchmarks show that UT2 improves both performance and stability over strong baselines, with larger gains as non-monotonicity and stochasticity increase.

Index Terms—Cooperative multi-agent reinforcement learning, credit assignment, value decomposition, individual-global max, target transformation.

I. INTRODUCTION

MULTI-AGENT reinforcement learning (MARL) studies how a team of agents learns coordinated behaviors through interaction with a shared environment. It provides a practical foundation for decision-making problems that require decentralized execution under partial observability and limited communication, and it has been increasingly adopted in real-world systems such as autonomous aerial vehicles [1], [2], smart

transportation [3], [4], medical image processing [5], [6], and resource optimization [7], [8].

In cooperative MARL, decentralized policies are commonly adopted due to communication constraints and the intractably large joint action space. A straightforward approach is independent learning, where each agent treats the others as part of the environment. However, simultaneous exploration typically induces a non-stationary learning problem and can substantially destabilize training. In simulated or laboratory settings, it is often possible to leverage additional information during learning, which can significantly improve policy optimization [9], [10], [11]. This motivates centralized training with decentralized execution (CTDE), where agents are trained with access to global information but executed based only on local action-observation histories [12], [13].

A major challenge under CTDE in cooperative settings is credit assignment, namely attributing a shared team reward to individual agents' decisions. A widely used solution is value decomposition, which represents the joint action-value as an aggregation of per-agent utilities to support decentralized greedy action selection. To ensure consistency between decentralized greedy action selection and centralized optimality [14], QMIX [15] assumes that the joint action-value is monotonic in per-agent utilities and learns a monotonic mixing function. This formulation is scalable and effective, but it also introduces a fundamental representational limitation. The restricted function class can be insufficient to fit non-monotonic learning targets, which may lead to convergence to suboptimal joint actions [16], [17], [18]. Many follow-up works extend QMIX along axes such as communication, exploration, and role-based learning [19], [20], [21], [22], [23], [24], [25], [26], [27], but they largely retain the monotonic mixing constraint and therefore inherit the same representational limitation.

Existing attempts to mitigate this limitation can be broadly grouped into two categories. The first category increases the representational capacity of value factorization, for example by introducing joint actions or pairwise interactions into the mixing structure [28], [29]. While effective in some settings, these approaches can be less scalable due to the increased computational burden of centralized learning as the number of agents grows. The second category biases learning so that monotonic approximators focus on potentially optimal joint actions.

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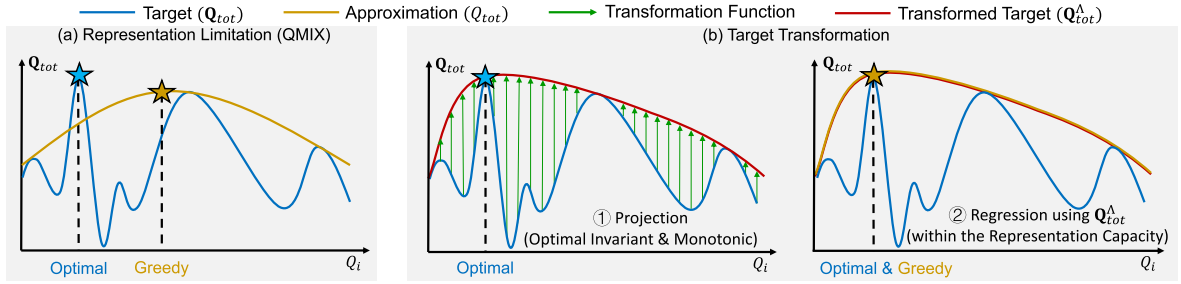


Fig. 1. The key idea of Target Transformation. (a) Monotonic value decomposition is constrained by limited representational capacity and may fail to recover the optimal policy when directly regressing non-monotonic or stochastic targets. (b) Target Transformation modifies suboptimal target values to obtain a monotonic representation space while preserving the optimal joint action. This enables monotonic value decomposition to fit the transformed targets and recover the optimal policy.

For example, WQMIX [17] down-weights joint actions whose learning targets fall below the current estimate. In stochastic environments, however, high returns can arise from rare lucky realizations rather than true optimality. This creates an intrinsic tension, where aggressive reweighting can overemphasize lucky but suboptimal samples, whereas conservative reweighting may fail to overcome representational limitation. We formalize this tension and show that reweighting alone can be insufficient under non-monotonic and stochastic targets.

To address these challenges, we propose *Target Transformation*, a theoretically grounded scheme for cooperative MARL with shared rewards. As illustrated in Fig. 1, our scheme reshapes non-monotonic and stochastic learning targets into a monotonic-representable surrogate for monotonic value decomposition while preserving the optimal joint action. Building on this idea, we introduce *Uncertainty-aware Target Transformation* (UT2), which combines uncertainty estimation, a best-individual coordination envelope, and a projection operator. Under a unique-optimum condition, the transformed targets preserve optimality and enable monotonic factorization to recover an optimal joint policy in the transformed target space, which coincides with the original optimum.

We further develop two practical algorithms as tractable instantiations of the theoretical procedure, named *Value-based UT2* (UT2-Q) and *Policy-based UT2* (UT2-P). We use a causal Transformer to model environment dynamics and reward autoregressively, and use its predictive variance as an uncertainty signal. We learn the best-individual coordination envelope via self-imitation learning, and define the transformation by replacing targets that are uncertain or not reliably underestimated with a conservative projection based on the minimum of best-individual values across agents. UT2-Q trains the joint value function by regressing to the projected targets, while UT2-P updates decentralized policies by maximizing a centralized critic trained on the same projected targets.

We evaluate UT2 on matrix games, Predator Prey (PP) [30], Predator Stag Hare (PSH), StarCraft Multi-Agent Challenge (SMAC) [31], SMACv2 [32], and Multi-Agent Coordination Challenge (MACO) [33]. Across these benchmarks, UT2 consistently improves performance and stability over representative baselines, with benefits becoming more pronounced as non-monotonicity and stochasticity increase. We further provide ablations to quantify the contribution of UT2 components and

discuss limitations when multiple optimal joint actions exist, together with practical mitigation strategies.

II. PRELIMINARIES

This section defines the cooperative MARL problem under the Decentralized Partially Observable Markov Decision Process (Dec-POMDP) and reviews value decomposition.

A. Dec-Pomdp

A fully cooperative MARL problem is often described as a Dec-POMDP [34], consisting of a tuple $G = \langle S, U, P, r, Z, O, n, \gamma \rangle$. Here $s \in S$ denotes the global state. At each timestep, each agent $a \in A \equiv \{1, 2, \dots, n\}$ selects an action $u_a \in U$, forming a joint action $\mathbf{u} \in U^n$. The environment transitions according to $P(s' | s, \mathbf{u})$ and emits a shared team reward $r(s, \mathbf{u})$, with discount factor $\gamma \in [0, 1]$.

Due to partial observability, each agent receives an individual observation $z \in Z$ produced by the observation function $O(s, a) : S \times A \rightarrow Z$. Each agent maintains an action-observation history $\tau_a \in T \equiv (Z \times U)^*$ and conditions a potentially stochastic policy $\pi_a(u_a | \tau_a) : T \times U \rightarrow [0, 1]$. Let $\tau = (\tau_1, \dots, \tau_n)$ denote the joint history, and let the joint policy be $\pi = \{\pi_a\}_{a=1}^n$.

The joint policy π induces a joint action-value function $Q^\pi(s_t, \mathbf{u}_t) = \mathbb{E}_{s_{t+1:\infty}, \mathbf{u}_{t+1:\infty} \sim \pi} [G_t | s_t, \mathbf{u}_t]$, where the discounted return is $G_t = \sum_{i=0}^{\infty} \gamma^i r_{t+i}$. The optimal joint action-value function $Q^*(s, \mathbf{u}) = \max_{\pi} Q^\pi(s, \mathbf{u})$ and the *optimal joint action* at state s as $\mathbf{u}^*(s) \in \arg \max_{\mathbf{u} \in U} Q^*(s, \mathbf{u})$.

B. Value Decomposition

Under centralized training with decentralized execution, agents execute policies conditioned only on local histories τ_a but can be trained with global information such as s and τ . Value decomposition introduces a mixing function to combine individual utilities into a centralized joint value function learned by monotonic value decomposition methods,

$$Q_{tot}(s, \tau, \mathbf{u}; \theta) = f(s, \{Q_a(\tau_a, u_a; \theta_q)\}_{a=1}^n; \theta_f), \quad (1)$$

where $Q_a(\tau_a, u_a; \theta_q)$ is the per-agent utility network and f is the mixing network. The learnable parameters $\theta = (\theta_f, \theta_q)$ are updated by minimizing the difference temporal-difference (TD)

regression loss:

$$\mathcal{L}_{TD}(\theta) = \mathbb{E}_{\mathcal{D}}[\mathbf{Q}_{tot}(s, \mathbf{u}) - Q_{tot}(s, \boldsymbol{\tau}, \mathbf{u}; \theta)]^2, \quad (2)$$

$$\mathbf{Q}_{tot}(s, \mathbf{u}) = r(s, \mathbf{u}) + \gamma \max_{\mathbf{u}'} Q_{tot}(s', \boldsymbol{\tau}', \mathbf{u}'; \theta'), \quad (3)$$

where $\mathcal{D}$ is the replay buffer, θ' are the parameters of the target network $Q_{tot}(s', \boldsymbol{\tau}', \mathbf{u}'; \theta')$. In this work, Q_{tot} denotes the fitted function approximator, while $\mathbf{Q}_{tot}$ denotes the TD target value used to train Q_{tot} .

A significant challenge in value decomposition is to achieve the Individual-Global Max (IGM) principle [14]:

$$\mathbf{u}^g = \arg \max_{\mathbf{u}} Q_{tot}(s, \boldsymbol{\tau}, \mathbf{u}) = \begin{pmatrix} \arg \max_{u_1} Q_1(\tau_1, u_1) \\ \vdots \\ \arg \max_{u_n} Q_n(\tau_n, u_n) \end{pmatrix}, \quad (4)$$

where $\mathbf{u}^g$ is the *greedy joint action* induced by the fitted utilities under decentralized execution.

The IGM principle affirms the consistency of global and local action selections, as well as the factorization relationship between the joint Q -value function Q_{tot} and the local Q -value function Q_a . To guarantee IGM, VDN [35], QMIX [15], and WQMIX [17] estimate Q_{tot} with specific mixing functions f . VDN uses linear value decomposition and factorizes Q_{tot} into a sum of individual Q -value functions. QMIX proposes monotonic value decomposition and applies a state-dependent monotonic mixing network to combine individual Q -value functions. The restricted spaces of all Q_{tot} that linear and monotonic value decomposition can represent are:

$$\mathcal{Q}^l := \left\{ Q_{tot} \mid Q_{tot} = \sum_{a=1}^n Q_a(\tau_a, u_a) \right\}, \quad (5)$$

$$\mathcal{Q}^m := \{ Q_{tot} \mid Q_{tot} = f_s(Q_1(\tau_1, u_1), \dots, Q_n(\tau_n, u_n)) \}, \quad (6)$$

where f_s is parameterized as a feedforward network whose non-negative weights are generated by hypernetworks that take the state as input.

WQMIX introduces a weighting function into QMIX to bias TD regression toward potentially optimal joint actions:

$$\Pi_w Q := \arg \min_{Q_{tot} \in \mathcal{Q}^m} \mathbb{E}_{\mathcal{D}} w(s, \mathbf{u}) [\mathbf{Q}_{tot}(s, \mathbf{u}) - Q_{tot}(s, \boldsymbol{\tau}, \mathbf{u})]^2. \quad (7)$$

Since it is computationally infeasible to obtain the optimal joint action-value, WQMIX proposes two practical weighting rules, named Centrally-Weighted QMIX (CW-QMIX) and Optimistically-Weighted QMIX (OW-QMIX). Their weighting functions are:

$$w^{cw}(s, \mathbf{u}) = \begin{cases} 1 & \mathbf{Q}_{tot}(s, \mathbf{u}) > \hat{Q}_{tot}^*(s, \mathbf{u}^g) \text{ or } \mathbf{u} = \mathbf{u}^g \\ \alpha & \text{otherwise} \end{cases} \quad (8)$$

$$w^{ow}(s, \mathbf{u}) = \begin{cases} 1 & \mathbf{Q}_{tot}(s, \mathbf{u}) > Q_{tot}(s, \boldsymbol{\tau}, \mathbf{u}) \\ \alpha & \text{otherwise} \end{cases} \quad (9)$$

where $\alpha \in (0, 1]$, and $\hat{Q}_{tot}^*(s, \mathbf{u}^g)$ is an approximation of the optimal joint Q -value which is not constrained to be monotonic.

III. VALUE DECOMPOSITION UNDER NON-MONOTONIC AND STOCHASTIC LEARNING TARGETS

This section analyzes a key limitation of monotonic value decomposition when learning targets are *non-monotonic* and *stochastic*, with a focus on objectives that rely on a weighting function. To isolate the core phenomenon, we consider a simplified fully observed setting and omit the distinction between states and observations in the notation.

A. Limitations of Weighted Linear Value Decomposition

Direct analysis of WQMIX is complicated by the nonlinear monotonic mixing network. We therefore begin with a linear surrogate, weighted linear value decomposition (WLVD), to expose the essential limitation introduced by weighting. We construct the WLVD objective by modifying the WQMIX objective in (7):

$$\Pi_l Q := \arg \min_{Q_{tot} \in \mathcal{Q}^l} \mathbb{E}_{\mathcal{D}} w(s, \mathbf{u}) [\mathbf{Q}_{tot}(s, \mathbf{u}) - Q_{tot}(s, \mathbf{u})]^2, \quad (10)$$

where $w(s, \mathbf{u})$ is the optimistic weight defined in (9). Compared with WQMIX, the function class is changed from $\mathcal{Q}^m$ to $\mathcal{Q}^l$.

At a fixed state s , WLVD is a weighted least-squares projection of the target table $\mathbf{Q}_{tot}(s, \cdot)$ onto the additive subspace $\mathcal{Q}^l$. Consequently, changing $w(s, \mathbf{u})$ changes the projection geometry but does not change the representable class. Even in the deterministic regime where targets have no sampling variance, representational limitation can prevent recovering the optimal joint action. Consider Matrix Game A in Fig. 2(a) with $a = 8$, $b = -12$, and $c = 6$, and a uniformly distributed dataset with equal weights across joint actions. Solving the corresponding least-squares system yields additive utilities whose greedy joint action satisfies $\mathbf{u}^g \in \{(u_2, u_2), (u_2, u_3), (u_3, u_2), (u_3, u_3)\}$, while the optimal joint action is $\mathbf{u}^* = (u_1, u_1)$. Therefore, linear value decomposition can fail purely due to representational limitation, independent of stochasticity.

Weighting can partially mitigate this deterministic mismatch by de-emphasizing joint actions that are incompatible with additivity, which imposes an upper bound on α in optimistic weighting rules. In other words, to correct non-monotonic representational error, α must be sufficiently small so that the regression concentrates on potentially optimal regions.

When targets are stochastic, optimistic weighting introduces a competing requirement. The weight depends on realized targets and can selectively amplify lucky but suboptimal samples, which imposes a lower bound on α for stability. These constraints can conflict, which is formalized by Theorem 1.

Theorem 1: Let $\mathbf{u}^g$ be the greedy joint action induced by the solution of $\Pi_l Q$ in (10). There exists a class of target value functions $\mathbf{Q}_{tot}$ such that, for all $\alpha \in (0, 1]$, $\mathbf{u}^g \neq \mathbf{u}^*$.

Proof sketch: Fix a state s and an ϵ -greedy data distribution with behavior greedy joint action $\bar{\mathbf{u}} = (u_2, u_2)$. Under stochastic targets, the optimistic weight depends on realized values and can amplify lucky high outcomes of a suboptimal joint action. This yields a stability requirement that α cannot be too small.

(a) Matrix Game A				(b) Matrix Game B			
	u_1	u_2	u_3		u_1	u_2	u_3
u_1	a	b	b	u_1	a	b	b
u_2	b	c	c	u_2	b	c/d	e
u_3	b	c	c	u_3	b	e	e
(c) OW-QMIX($\alpha = 0.5$)				(d) CW-QMIX($\alpha = 0.5$)			
	u_1	u_2	u_3		u_1	u_2	u_3
u_1	7.34	7.29	6.06	u_1	7.33	7.25	6.33
u_2	7.29	7.23	6.05	u_2	7.25	7.2	6.28
u_3	6.05	6.03	5.92	u_3	6.33	6.28	5.45
(e) OW-QMIX($\alpha = 0.1$)				(f) CW-QMIX($\alpha = 0.1$)			
	u_1	u_2	u_3		u_1	u_2	u_3
u_1	7.29	7.67	6.42	u_1	7.98	7.98	6.33
u_2	7.67	10.9	6.62	u_2	7.98	7.98	6.33
u_3	6.43	6.63	5.73	u_3	6.33	6.33	5.51

Fig. 2. The payoff of matrix games and the approximated Q_{tot} returned by WQMIX on Matrix Game B. Matrix Game A is deterministic and non-monotonic in the sense that the target ordering cannot be represented by monotonic factorization without de-emphasizing some suboptimal joint actions. Matrix Game B is stochastic due to a two-point payoff at (u_2, u_2) , where $Q_{tot}(u_2, u_2) = c$ with probability p and d with probability $1 - p$. Boldface indicates the optimal or greedy joint action.

We now give a concrete counterexample that combines non-monotonicity and stochasticity. Using Fig. 2(b) structure with $a = 8$, $b = -12$, $c = 12$, $d = 0$, $e = 6$, and $p = 0.5$, for $\epsilon \in \{1, 0.75, 0.5, 0.25\}$ there does not exist any $\alpha \in (0, 1]$ such that the WLVD solution induces $\mathbf{u}^g = \mathbf{u}^*$. Hence no single optimistic weight choice can simultaneously suppress stochastic optimistic amplification and overcome representational limitation. $\square$

Theorem 1 reveals an intrinsic tension in stochastic environments. To mitigate representational limitation under non-monotonic targets, α must be small enough to downweight incompatible regions. To avoid overestimating lucky suboptimal samples under stochastic targets, α must be large enough to prevent selective amplification. As a result, weighting can reshape the effective learning distribution in a way that leads to incorrect greedy selection when targets are both non-monotonic and stochastic.

B. Limitations of Weighted Monotonic Value Decomposition

We now show that WQMIX exhibits the same qualitative tension as WLVD. Consider Matrix Game B in Fig. 2(b) with $a = 8$, $b = e = 6$, $c = 12$, $d = 0$, and $p = 0.5$, and assume $\mathcal{D}$ is uniformly distributed. The expected target ordering is monotonic, but realized targets are stochastic due to the random payoff at (u_2, u_2) . Let the suboptimal joint action be $\mathbf{u}^s = (u_2, u_2)$ and the optimal joint action be $\mathbf{u}^* = (u_1, u_1)$. When α is relatively small, optimistic weighting repeatedly emphasizes lucky realizations in which $Q_{tot}(\mathbf{u}^s)$ takes the high value c . This can inflate the fitted value of $\mathbf{u}^s$ and flip the learned $\arg \max$. OW-QMIX can exacerbate this effect due to more aggressive optimistic

weighting. CW-QMIX can further oscillate when $\mathbf{u}^s$ alternates between greedy and non-greedy as stochastic targets switch between high and low realizations, preventing convergence to a stable greedy joint action.

Taken together, these observations suggest that *reweighting alone* is not a robust mechanism for resolving representational limitation under non-monotonic and stochastic learning targets. The same weight parameter must satisfy conflicting requirements. It must be sufficiently small to address representational mismatch in the non-monotonic regime, but sufficiently large to avoid optimistic amplification in the stochastic regime.

IV. METHODOLOGY

This section introduces *Target Transformation*, a new operator for cooperative value decomposition. It addresses the representational limitation induced by monotonic mixing by reshaping learning targets while preserving the optimal joint action.

A. Target Transformation

Monotonic value decomposition methods can only represent joint action-values in the monotonic function class $\mathcal{Q}^m$ in (5). Our key idea is to transform the target joint action-values Q_{tot} into a monotonic-representable surrogate while keeping the optimal joint action unchanged in expectation. When rewards or transitions are stochastic, $Q_{tot}(s, \mathbf{u})$ is a random variable and $\mathbb{E}[Q_{tot}(s, \mathbf{u})]$ denotes its expectation.

Definition 1 (Target Transformation): A *Target Transformation* is an operator Λ^* that maps a target joint action-value function Q_{tot} to a transformed target Λ^*Q_{tot} such that, for any fixed state s , $\Lambda^*Q_{tot}(s, \cdot) \in \mathcal{Q}^m$, and $\arg \max_{\mathbf{u}} \mathbb{E}[\Lambda^*Q_{tot}(s, \mathbf{u})] = \arg \max_{\mathbf{u}} \mathbb{E}[Q_{tot}(s, \mathbf{u})]$.

Let $\mathcal{M}$ denote the Markov Decision Process (MDP) with target values Q_{tot} , and let $\mathcal{M}_{\Lambda^*}$ denote the transformed MDP induced by Λ^*Q_{tot} . Definition 1 formalizes two desired properties:

- 1) *Monotonic-representable Transformed Target:* All target value after the projection $\Lambda^*Q_{tot} \in \mathcal{Q}^m$.
- 2) *Optimal-policy Invariance:* $\mathcal{M}$ and $\mathcal{M}_{\Lambda^*}$ share the same optimal joint action at each state.

If both properties hold, then monotonic value decomposition can fit the transformed target and recover an optimal joint policy of the original MDP.

In the extreme, if we only require the ordering of joint actions rather than the exact values of suboptimal joint actions, we may keep the optimal target unchanged and replace all other targets with any values strictly below the optimal one, as long as the resulting target is monotonic-representable

$$\Lambda^*Q_{tot}(s, \mathbf{u}) = \begin{cases} Q_{tot}(s, \mathbf{u}), & \mathbf{u} = \mathbf{u}^*(s), \\ Q_{tot}^f(s, \mathbf{u}), & \text{otherwise,} \end{cases} \quad (11)$$

where $Q_{tot}^f(s, \mathbf{u}) < Q_{tot}(s, \mathbf{u}^*(s))$ is chosen so that $\mathbb{E}[\Lambda^*Q_{tot}(s, \cdot)] \in \mathcal{Q}^m$. This highlights the principle that recovering the optimal policy only requires preserving the optimal joint action, rather than regressing exact suboptimal values.

We now provide a concrete instance of Target Transformation. First, we introduce two auxiliary quantities, including an uncertainty indicator $\mathbb{I}(s, \mathbf{u})$ identifies whether the target at $(s, \mathbf{u})$ is stochastic, as well as the best-individual value function $\Omega_a(s, u_a)$ upper-bounds the expected return achievable by fixing agent a 's action and allowing the remaining agents to respond optimally in expectation

$$\mathbb{I}(s, \mathbf{u}) \triangleq \begin{cases} 1, & \mathbf{Q}_{tot}(s, \mathbf{u}) \text{ is stochastic} \\ 0, & \text{otherwise,} \end{cases}$$

$$\Omega_a(s, u_a) \triangleq \max_{\mathbf{u}_{-a}} \mathbb{E}[\mathbf{Q}_{tot}(s, u_a, \mathbf{u}_{-a})],$$

where $\mathbf{u} = (u_a, \mathbf{u}_{-a})$ and $\mathbf{u}_{-a}$ denotes the joint action of all agents except a . The maximization removes any dependence on the concrete $\mathbf{u}_{-a}$, hence $\Omega_a(s, u_a)$ depends only on (s, u_a) .

Using these quantities, we define the projected target $\mathbf{Q}_{tot}^\Lambda$:

$$\mathbf{Q}_{tot}^\Lambda(s, \mathbf{u}) = \begin{cases} \mathbf{Q}_{tot}(s, \mathbf{u}), & \mathbb{I}(s, \mathbf{u}) = 0 \text{ and } \delta(s, \mathbf{u}) > 0 \\ \min \Omega_a(s, u_a), & \text{otherwise,} \end{cases} \quad (12)$$

where $\delta(s, \mathbf{u}) = \mathbf{Q}_{tot}(s, \mathbf{u}) - Q_{tot}(s, \mathbf{u})$ measures the current underestimation of the target by the learned critic. The rule has two cases. When a target is *deterministic* and *underestimated*, i.e., $\mathbb{I}(s, \mathbf{u}) = 0$ and $\delta(s, \mathbf{u}) > 0$, we keep the original target value unchanged. This preserves optimistic but reliable learning signals for potentially optimal actions. In all other cases, we replace the target by $\min_a \Omega_a(s, u_a)$, which acts as a monotonic surrogate that encodes suboptimality without requiring the exact non-monotonic target surface to be representable.

The use of the min operator is essential for policy preservation. $\Omega_a(s, u_a)$ is an upper envelope over the responses of the other agents given u_a . If one aggregates these envelopes with $\max_a$ instead, then any joint action that contains an individually optimal component can attain the same projected value as the true optimum. This introduces additional maximizers in the transformed problem and can break *optimal-policy invariance* under monotonic value decomposition. In contrast, $\min_a \Omega_a(s, u_a)$ yields a conservative aggregation that maintains a strict separation between $\mathbf{u}^*(s)$ and any $\mathbf{u} \neq \mathbf{u}^*(s)$ under the following standard assumption.

Assumption 1 (Unique Optimal Joint Policy): At each given state $s \in S$, the optimal joint action $\mathbf{u}^*(s)$ is unique.

Theorem 2: Under Assumption 1, the projected target $\mathbf{Q}_{tot}^\Lambda$ defined in (12) is a Target Transformation in the sense of Definition 1.

Proof: (a) *Optimal-policy Invariance:* Fix a state s and let $\mathbf{u}^* \triangleq \arg \max_{\mathbf{u}} \mathbb{E}[\mathbf{Q}_{tot}(s, \mathbf{u})]$. For any joint action $\mathbf{u} = (u_a, \mathbf{u}_{-a})$ and any agent a , feasibility of $\mathbf{u}_{-a}$ in the maximization gives $\mathbb{E}[\mathbf{Q}_{tot}(s, \mathbf{u})] \leq \max_{\mathbf{u}'_{-a}} \mathbb{E}[\mathbf{Q}_{tot}(s, u_a, \mathbf{u}'_{-a})] = \Omega_a(s, u_a)$. Taking $\min_a$ preserves the inequality and yields $\mathbb{E}[\mathbf{Q}_{tot}(s, \mathbf{u})] \leq \min_a \Omega_a(s, u_a)$.

We evaluate the transformed value at $\mathbf{u}^*$. If $\mathbf{Q}_{tot}^\Lambda(s, \mathbf{u}^*) = \mathbf{Q}_{tot}(s, \mathbf{u}^*)$, then $\mathbb{E}[\mathbf{Q}_{tot}^\Lambda(s, \mathbf{u}^*)] = \mathbb{E}[\mathbf{Q}_{tot}(s, \mathbf{u}^*)]$. Otherwise $\mathbf{Q}_{tot}^\Lambda(s, \mathbf{u}^*) = \min_a \Omega_a(s, u_a^*)$. For each agent a , choosing $\mathbf{u}_{-a} = \mathbf{u}_{-a}^*$ is feasible, hence $\Omega_a(s, u_a^*) \geq \mathbb{E}[\mathbf{Q}_{tot}(s, \mathbf{u}^*)]$, and

equality holds because $\mathbf{u}^*$ is optimal in expectation. Therefore $\min_a \Omega_a(s, u_a^*) = \mathbb{E}[\mathbf{Q}_{tot}(s, \mathbf{u}^*)]$, and in both branches $\mathbb{E}[\mathbf{Q}_{tot}^\Lambda(s, \mathbf{u}^*)] = \mathbb{E}[\mathbf{Q}_{tot}(s, \mathbf{u}^*)]$.

Now consider any $\mathbf{u} \neq \mathbf{u}^*$. By Assumption 1 there exists an agent a with $u_a \neq u_a^*$. Suppose $\Omega_a(s, u_a) \geq \mathbb{E}[\mathbf{Q}_{tot}(s, \mathbf{u}^*)]$. Then there exists $\tilde{\mathbf{u}}_{-a}$ such that $\mathbb{E}[\mathbf{Q}_{tot}(s, u_a, \tilde{\mathbf{u}}_{-a})] \geq \mathbb{E}[\mathbf{Q}_{tot}(s, \mathbf{u}^*)]$. Let $\tilde{\mathbf{u}} = (u_a, \tilde{\mathbf{u}}_{-a})$. Since $u_a \neq u_a^*$, we have $\tilde{\mathbf{u}} \neq \mathbf{u}^*$, which contradicts the uniqueness of $\mathbf{u}^*$. Therefore $\Omega_a(s, u_a) < \mathbb{E}[\mathbf{Q}_{tot}(s, \mathbf{u}^*)]$.

Since $\mathbf{u} \neq \mathbf{u}^*$ implies $\mathbb{E}[\mathbf{Q}_{tot}^\Lambda(s, \mathbf{u})] = \min_i \Omega_i(s, u_i)$, we obtain $\mathbb{E}[\mathbf{Q}_{tot}^\Lambda(s, \mathbf{u})] = \min_i \Omega_i(s, u_i) \leq \Omega_a(s, u_a) < \mathbb{E}[\mathbf{Q}_{tot}(s, \mathbf{u}^*)] = \mathbb{E}[\mathbf{Q}_{tot}^\Lambda(s, \mathbf{u}^*)]$. Hence

$$\arg \max_{\mathbf{u}} \mathbb{E}[\mathbf{Q}_{tot}^\Lambda(s, \mathbf{u})] = \mathbf{u}^*. \quad (13)$$

(b) *Monotonic-representable Transformed Target:* We next show that the transformed target $\mathbf{Q}_{tot}^\Lambda$ satisfies the monotonicity requirement of $\mathcal{Q}^m$. Equivalently, for any agent a , if there exist $u_a \neq u'_a$ and some $\mathbf{u}_{-a}$ such that $\mathbb{E}[\mathbf{Q}_{tot}^\Lambda(s, u_a, \mathbf{u}_{-a})] > \mathbb{E}[\mathbf{Q}_{tot}^\Lambda(s, u'_a, \mathbf{u}_{-a})]$, then this preference ordering is preserved for any alternative $\mathbf{u}'_{-a}$, i.e., $\mathbb{E}[\mathbf{Q}_{tot}^\Lambda(s, u_a, \mathbf{u}'_{-a})] \geq \mathbb{E}[\mathbf{Q}_{tot}^\Lambda(s, u'_a, \mathbf{u}'_{-a})]$. The following chain of implications verifies this property directly:

$$\begin{aligned} & \mathbb{E}[\mathbf{Q}_{tot}^\Lambda(s, u_a, \mathbf{u}_{-a})] > \mathbb{E}[\mathbf{Q}_{tot}^\Lambda(s, u'_a, \mathbf{u}_{-a})] \\ \implies & \min_{\mathbf{u}'_{-a}} [\max_{\mathbf{u}_{-a}} \mathbb{E}[\mathbf{Q}_{tot}(s, u_a, \mathbf{u}_{-a})], \max_{\mathbf{u}_{-a}} \mathbb{E}[\mathbf{Q}_{tot}(s, u'_a, \mathbf{u}_{-a})]] \\ & > \min_{\mathbf{u}'_{-a}} [\max_{\mathbf{u}_{-a}} \mathbb{E}[\mathbf{Q}_{tot}(s, u'_a, \mathbf{u}_{-a})], \max_{\mathbf{u}_{-a}} \mathbb{E}[\mathbf{Q}_{tot}(s, u_a, \mathbf{u}_{-a})]] \\ \implies & \max_{\mathbf{u}_{-a}} \mathbb{E}[\mathbf{Q}_{tot}(s, u_a, \mathbf{u}_{-a})] \geq \mathbb{E}[\max_{\mathbf{u}_{-a}} \mathbf{Q}_{tot}(s, u'_a, \mathbf{u}_{-a})] \\ \implies & \min_{\mathbf{u}'_{-a}} [\max_{\mathbf{u}_{-a}} \mathbb{E}[\mathbf{Q}_{tot}(s, u_a, \mathbf{u}_{-a})], \max_{\mathbf{u}_{-a}} \mathbb{E}[\mathbf{Q}_{tot}(s, u_a, \mathbf{u}'_{-a})]] \\ & \geq \min_{\mathbf{u}'_{-a}} [\max_{\mathbf{u}_{-a}} \mathbb{E}[\mathbf{Q}_{tot}(s, u'_a, \mathbf{u}_{-a})], \max_{\mathbf{u}_{-a}} \mathbb{E}[\mathbf{Q}_{tot}(s, u_a, \mathbf{u}'_{-a})]] \\ \iff & \mathbb{E}[\mathbf{Q}_{tot}^\Lambda(s, u_a, \mathbf{u}'_{-a})] \geq \mathbb{E}[\mathbf{Q}_{tot}^\Lambda(s, u'_a, \mathbf{u}'_{-a})]. \end{aligned}$$

Hence, $\mathbf{Q}_{tot}^\Lambda$ is monotonic-representable, i.e., $\mathbf{Q}_{tot}^\Lambda \in \mathcal{Q}^m$.

Combining (a) and (b), the projection in (12) satisfies both requirements in Definition 1 and is therefore a Target Transformation. $\square$

Fig. 3 contrasts our projection with two natural alternatives obtained by replacing Ω_a with counterfactual- or best-response-style quantities. For $\mathbf{u} = (u_a, \mathbf{u}_{-a})$, define

$$\Omega_a^{CF}(s, \mathbf{u}) \triangleq \mathbb{E} \left[\mathbf{Q}_{tot}(s, \mathbf{u}) - \sum_{u'_a} \bar{P}_{u'_a} \mathbf{Q}_{tot}(s, u'_a, \mathbf{u}_{-a}) \right],$$

$$\Omega_a^{BR}(s, \mathbf{u}) \triangleq \max_{u'_a} \mathbb{E}[\mathbf{Q}_{tot}(s, u'_a, \mathbf{u}_{-a})],$$

where $\bar{P}_{u'_a} = P(u'_a | s, \mathbf{u}_{-a})$. Counterfactual constructions depend on $\mathbf{u}_{-a}$ and can invert the ordering between two actions across different $\mathbf{u}_{-a}$, which violates monotonic representability. Best-response constructions can introduce multiple maximizers under max-type aggregation, which breaks optimal-policy invariance. In contrast, $\min_a \Omega_a(s, u_a)$ avoids both failure modes under Assumption 1.

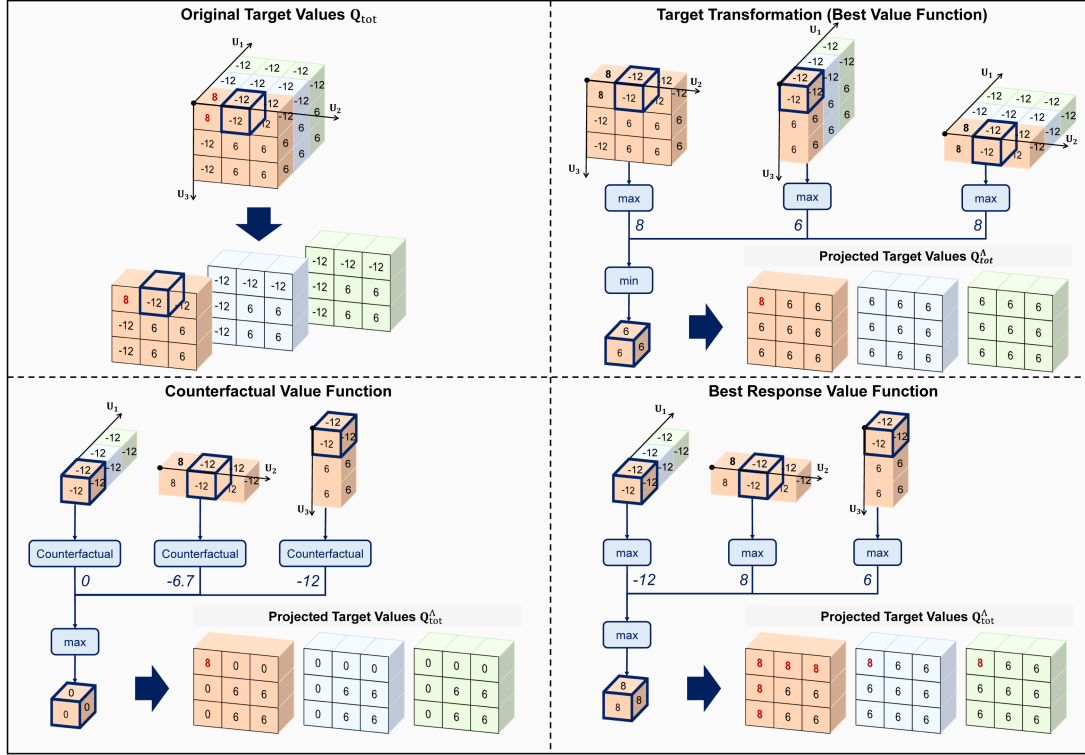


Fig. 3. The projection process and results for the target value functions using the best-individual value function, a counterfactual construction, and a best-response construction.

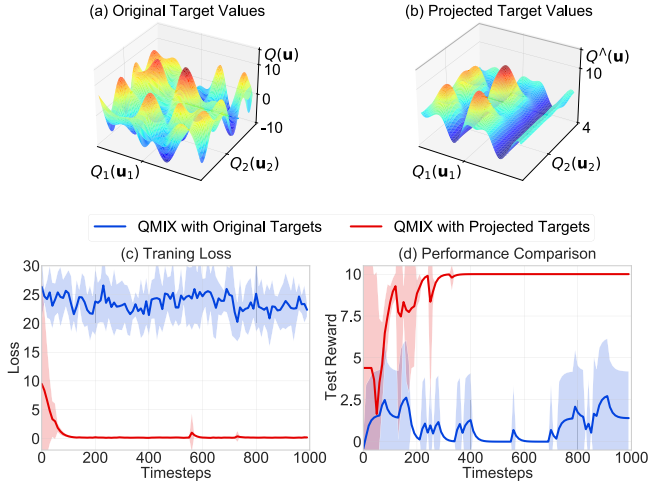


Fig. 4. Visualization of the projected target values, the training loss, and the results using the original and the projected targets.

Fig. 4(a) presents an illustrative 10×10 matrix game target $Q_{tot}(s, \mathbf{u})$ with a unique optimal value at $(\mathbf{u}_5, \mathbf{u}_5) = 10$, while the remaining entries are uniformly sampled in $[-10, 9]$. Fig. 4(b) shows the transformed target $Q_{tot}^A(s, \mathbf{u})$. Fig. 4(c) and (d) compare monotonic value decomposition trained on the original targets and on the transformed targets. Directly fitting Q_{tot} yields persistent regression error due to representational limitation and can fail to recover the optimal joint action. In contrast, fitting Q_{tot}^A yields near-zero regression error and converges to the optimal joint action.

An immediate question is why UT2 does not converge to the remaining suboptimal entries in Fig. 4. UT2 does not aim to fit the original non-monotonic surface everywhere. UT2 constructs Q_{tot}^A so that $\mathbf{u}^*(s)$ remains the unique maximizer of $\mathbb{E}[Q_{tot}^A(s, \cdot)]$ under Assumption 1. Once the transformed target is representable by monotonic mixing, the greedy joint action induced by monotonic value decomposition recovers $\mathbf{u}^*(s)$ and avoids spurious attractors caused by irreducible regression error on Q_{tot} .

B. Uncertainty of Value Function

Inspired by the analysis in RND [36], uncertainty in multi-agent value learning can arise from limited data, model misspecification, or optimization dynamics. In this work, UT2 focuses on environment stochasticity, which is aleatoric uncertainty. This is the regime in which reweighting-based methods can overemphasize lucky high-return samples and become unstable.

To estimate aleatoric uncertainty, we fit a heteroscedastic diagonal Gaussian distribution to next-state prediction and reward prediction. We predict the mean of the next state $\hat{s}_{t+1}$ and the mean reward $\hat{r}_t$ together with diagonal covariances $\hat{\Sigma}_{t+1}^s$ and $\hat{\Sigma}_t^r$. Using heteroscedastic maximum likelihood [37], [38], we minimize

$$\mathcal{L} = (s_{t+1} - \hat{s}_{t+1})^\top \hat{\Sigma}_{t+1}^{s,-1} (s_{t+1} - \hat{s}_{t+1}) + \lambda \log(\det(\hat{\Sigma}_{t+1}^s)) + (r_t - \hat{r}_t)^\top \hat{\Sigma}_t^{r,-1} (r_t - \hat{r}_t) + \lambda \log(\det(\hat{\Sigma}_t^r)), \quad (14)$$

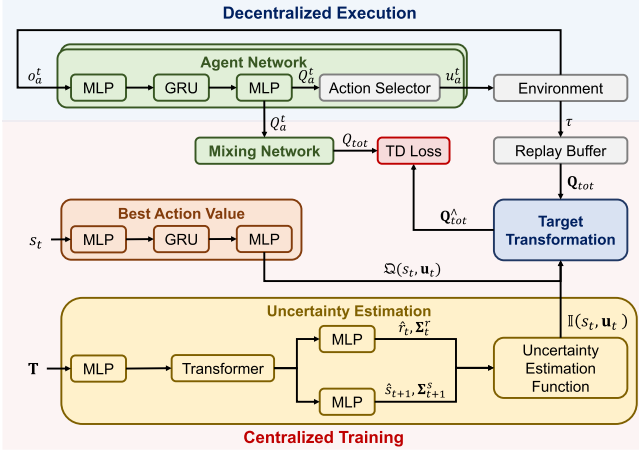


Fig. 5. An overview of UT2.

where the quadratic terms are scaled by the predicted uncertainty, the log-determinant terms regularize uncertainty magnitude, and λ controls the aleatoric uncertainty budget. We parameterize $\log \Sigma$ for stable optimization.

Finally, we set thresholds for $\hat{\Sigma}_{t+1}^s$ and $\hat{\Sigma}_t^r$. If the predicted uncertainty for a given $(s, \mathbf{u})$ exceeds these thresholds, we set $\mathbb{I}(s, \mathbf{u}) = 1$, otherwise we set $\mathbb{I}(s, \mathbf{u}) = 0$. This indicator is then used by (12) to decide whether to apply the conservative target projection in UT2.

V. PRACTICAL IMPLEMENTATION

The analysis above characterizes Target Transformation in an idealized setting. We now instantiate it as scalable deep MARL algorithms under CTDE with neural function approximation. We present two implementations. A value-based variant UT2-Q targets discrete action spaces. A policy-based variant UT2-P targets continuous or large action spaces. Both implementations share three ingredients. An uncertainty indicator $\mathbb{I}(s, \mathbf{u})$ detects environmental stochasticity. A best-individual value estimator approximates the coordination envelope $\bar{\mathbf{Q}}_a$. A projected TD target $\mathbf{Q}_{tot}^A$ is constructed to train a joint value function learned by monotonic value decomposition methods.

A. Uncertainty Estimation Function

We estimate whether a given state and joint-action pair $(s_t, \mathbf{u}_t)$ exhibits substantial aleatoric stochasticity by predicting the next state and reward together with heteroscedastic predictive uncertainty. Concretely, we train a world model $\mathbf{WM}$ on trajectories

$$\mathbf{T} = \{\dots, s_t, u_t^1, \dots, u_t^n, r_t, s_{t+1}, \dots\}. \quad (15)$$

The model $\mathbf{WM}$ consists of a causal Transformer f_c followed by two prediction heads. A transition head f_m predicts the mean next state and its diagonal standard deviation. A reward head f_r predicts the mean reward and its diagonal standard deviation

$$\text{Causal Transformer: } f_c(x | \mathbf{T}_{<x}; \psi_c), \quad (16)$$

$$\text{Transition Head: } f_m(\hat{s}_{t+1}, \hat{\Sigma}_t^s | f_c(s_t | \mathbf{T}_{<s_t}); \psi_m), \quad (17)$$

$$\text{Reward Head: } f_r(\hat{r}_t, \hat{\Sigma}_t^r | f_c(r_t | \mathbf{T}_{<r_t}); \psi_r). \quad (18)$$

We fit a heteroscedastic Gaussian by minimizing the negative log-likelihood:

$$\begin{aligned} \mathcal{L}(\psi_c, \psi_m, \psi_r) = & \frac{1}{2T} \sum_{t=0}^{T-1} \left[\frac{(r_t - \hat{r}_t)^2}{(\hat{\Sigma}_t^r)^2} + \log(\hat{\Sigma}_t^r)^2 \right. \\ & \left. + \frac{\|s_{t+1} - \hat{s}_{t+1}\|^2}{(\hat{\Sigma}_t^s)^2} + \log(\hat{\Sigma}_t^s)^2 \right], \quad (19) \end{aligned}$$

where larger predicted variance indicates higher transition or reward stochasticity for that input.

We define the uncertainty indicator:

$$\mathbb{I}(s_t, \mathbf{u}_t) = \begin{cases} 1, & \hat{\Sigma}_t^r > r_b \text{ or } \hat{\Sigma}_t^s > s_b, \\ 0, & \text{otherwise,} \end{cases} \quad (20)$$

with thresholds r_b and s_b . Importantly, $\mathbb{I}(s, \mathbf{u})$ targets environmental stochasticity rather than epistemic uncertainty due to limited data.

B. Value-Based Uncertainty-Aware Target Transformation

We instantiate Target Transformation for value-based MARL in Dec-POMDPs. UT2-Q contains three trainable modules: 1) a QMIX-style joint value function Q_{tot} is used for decentralized execution, 2) a best-individual estimator $\bar{\mathbf{Q}}_a$ approximating $\bar{\mathbf{Q}}_a$, and 3) a projected TD target $\mathbf{Q}_{tot}^A$ is used to train Q_{tot} .

a) We adopt the standard monotonic factorization:

$$Q_{tot}(s, \tau, \mathbf{u}; \theta_f, \theta_q) = f(s, \{Q_a(\tau_a, u_a; \theta_q)\}_{a=1}^n; \theta_f), \quad (21)$$

where each Q_a is a deep recurrent Q -network (DRQN), and f is a mixing network with non-negative weights to ensure $\partial Q_{tot} / \partial Q_a \geq 0$. We train Q_{tot} by regressing to the projected target:

$$\mathcal{L}(\theta_f, \theta_q) = \mathbb{E}_{\tau \in b} [Q_{tot}(s, \tau, \mathbf{u}; \theta_f, \theta_q) - \mathbf{Q}_{tot}^A(s, \mathbf{u})]^2, \quad (22)$$

where b is a minibatch sampled from the replay buffer $\mathcal{D}$ and τ denotes the corresponding trajectory segment.

b) The ideal best-individual value is $\bar{\mathbf{Q}}_a(s, u_a) = \max_{\mathbf{u}_{-a}} \mathbb{E}[\mathbf{Q}_{tot}(s, u_a, \mathbf{u}_{-a})]$. Enumerating $\mathbf{u}_{-a}$ is computationally prohibitive when the number of agents or the action space is large. We therefore approximate it with a state-conditioned estimator $\tilde{Q}_a(s, u_a; \theta_w) \approx \bar{\mathbf{Q}}_a(s, u_a)$, which shares the same backbone as Q_a but is additionally conditioned on s since it is used only during centralized training.

Intuitively, $\tilde{Q}_a$ should capture how good action u_a can be when the other agents coordinate optimally with agent a in expectation. We learn $\tilde{Q}_a$ via a self-imitation objective that emphasizes experiences whose returns exceed the current envelope estimate and therefore provide a constructive improvement signal. We construct the discounted return target using the predicted mean rewards $\hat{r}_k$ produced by $\mathbf{WM}$ as $R_t = \sum_{k=t}^T \gamma^{k-t} \hat{r}_k$, and

optimize

$$\begin{aligned} \mathcal{L}(\theta_w) = & \mathbb{E}_{\tau \in m} \frac{\beta}{2} \left\| \mathbb{I}(s, \mathbf{u}) \left[R_t - \tilde{Q}_a(s, u_a; \theta_w) \right]_+ \right\|^2 \\ & + \mathbb{E}_{\tau \in b} \frac{1}{2} \left\| [1 - \mathbb{I}(s, \mathbf{u})] \left[R_t - \tilde{Q}_a(s, u_a; \theta_w) \right] \right\|^2, \end{aligned} \quad (23)$$

where $[x]_+ = \max\{x, 0\}$, $\beta > 1$ increases the contribution of informative samples flagged by the uncertainty indicator, m is a minibatch from a prioritized sampler, and b is a standard minibatch.

We use predicted mean rewards from the world model for two reasons. The envelope $\tilde{Q}_a(s, u_a)$ approximates an expectation-level best-coordination return under environment randomness. Equation (23) emphasizes large positive residuals, hence raw rewards can cause rare lucky but suboptimal realizations to induce spuriously large R_t . This biases $\tilde{Q}_a$ upward and miscalibrates the downstream projection $\min_a \tilde{Q}_a$ that forms $\mathbf{Q}_{tot}^\Lambda$. Predicted mean rewards reduce variance in the imitation signal, decouple the improvement weight from luck, and stabilize the envelope calibration used by target projection.

To implement the prioritized sampler in (23) and improve sample efficiency, we use prioritized replay for learning $\tilde{Q}_a$. We define the sampling priority

$$sp(s, u_a) = \left[R_t - \tilde{Q}_a(s, u_a; \theta_w) \right]_+. \quad (24)$$

Sampling proportionally to $sp(s, u_a)$ increases the fraction of transitions with non-zero imitation gradients, reduces data waste, and keeps all transitions eligible, which mitigates collapse to a narrow subset of the replay distribution. We use importance-sampling weights to correct the bias induced by non-uniform sampling, which reduces the risk of overestimation when prioritization concentrates on a small portion of the data.

The priority in (24) is an improvement signal for learning the best-individual envelope. It is not a Bellman temporal-difference error used for fixed-point evaluation in prioritized experience replay [39]. A good state-action pair for $\tilde{Q}_a$ satisfies $R_t > \tilde{Q}_a(s, u_a; \theta_w)$, meaning that this sample provides evidence that the current envelope estimate can be increased for (s, u_a) under best coordination by the other agents. Prioritizing transitions by the temporal-difference error of a joint value function would instead bias sampling toward regions with large joint-value approximation error, which is not aligned with the envelope-learning objective and can overemphasize noisy or uninformative regions.

c) Given $\{\tilde{Q}_a\}_{a=1}^n$, we form the projected TD target as

$$\mathbf{Q}_{tot}^\Lambda(s, \mathbf{u}) = \begin{cases} \mathbf{Q}_{tot}(s, \mathbf{u}), & (s, \mathbf{u}) \in \mathcal{V}, \\ \min_a \tilde{Q}_a(s, u_a; \theta_w), & \text{otherwise.} \end{cases} \quad (25)$$

Here $\mathcal{V}$ collects targets that we keep unchanged:

$$\mathcal{V} \triangleq \{(s, \mathbf{u}) \mid \mathbb{I}(s, \mathbf{u}) = 0 \text{ and } \delta(s, \mathbf{u}) > 0\}, \quad (26)$$

where $\delta(s, \mathbf{u}) = \mathbf{Q}_{tot}(s, \mathbf{u}) - Q_{tot}(s, \tau, \mathbf{u}; \theta_f, \theta_q)$ is the empirical underestimation gap on the sampled TD target. UT2-Q keeps deterministic targets only when they are underestimated

and projects the remaining cases to the conservative surrogate $\min_a \tilde{Q}_a$.

The TD target is computed as

$$\mathbf{Q}_{tot}(s, \mathbf{u}) = r(s, \mathbf{u}) + \gamma \max_{\mathbf{u}'} Q_{tot}(s', \tau', \mathbf{u}'; \theta'_f, \theta'_q),$$

where (θ'_f, θ'_q) are the target network parameters periodically copied from (θ_f, θ_q) . Overall, $\mathbf{Q}_{tot}^\Lambda$ preserves trustworthy learning signals while preventing lucky but suboptimal targets from contaminating regression, thereby maintaining a well-calibrated projection that remains compatible with monotonic value decomposition.

C. Policy-Based Uncertainty-Aware Target Transformation

Many real-world cooperative tasks involve continuous or large action spaces, where UT2-Q is only applicable to discrete action spaces. We therefore extend Target Transformation to a policy-based setting and refer to the resulting method as UT2-P. UT2-P comprises four components: 1) a centralized but factored critic Q_{tot}^μ provides globally informed gradients, 2) decentralized actor(s) μ_a generate actions, 3) a best-individual critic $\tilde{Q}_a^\mu$ approximates $\mathbf{Q}_{tot}^\mu$, 4) a projected target $\mathbf{Q}_{tot}^{\mu, \Lambda}$ is used to train Q_{tot}^μ .

a) UT2-P employs a centralized but factored critic to retain scalability with the number of agents. We use monotonic value decomposition within the actor-critic framework:

$$Q_{tot}^\mu(s, \tau, \mathbf{u}; \phi_g, \phi_q) = g(s, \{Q_a^{\mu_a}(\tau_a, u_a; \phi_q)\}_{a=1}^n; \phi_g), \quad (27)$$

where $Q_a^{\mu_a}$ is an agent-wise critic and g is a monotonic mixing network. Architecturally, $Q_a^{\mu_a}$ and g follow the same design as Q_a and f in (21), except that $Q_a^{\mu_a}$ takes (τ_a, u_a) and outputs a scalar to keep the actor update differentiable. The centralized critic is trained by regressing to the projected target:

$$\mathcal{L}(\phi_g, \phi_q) = \mathbb{E}_{\mathcal{D}} \left[Q_{tot}^\mu(s, \tau, \mathbf{u}; \phi_g, \phi_q) - \mathbf{Q}_{tot}^{\mu, \Lambda}(s, \mathbf{u}) \right]^2. \quad (28)$$

b) The actor update depends on the action space. For continuous actions, we use the deterministic policy gradient with the centralized critic:

$$\nabla_{\phi_u} J(\boldsymbol{\mu}) = \mathbb{E}_B [\nabla_{\phi_u} \boldsymbol{\mu} \nabla_{\boldsymbol{\mu}} Q_{tot}^\mu(s, \mu_1(\tau_1), \dots, \mu_n(\tau_n))], \quad (29)$$

where $\boldsymbol{\mu} = \{\mu_1(\tau_1; \phi_u), \dots, \mu_n(\tau_n; \phi_u)\}$ and all agents share the actor parameters ϕ_u .

For discrete actions, we employ the Gumbel-Softmax reparameterization to obtain a differentiable surrogate action $\mathbf{x} = \{x_1, \dots, x_n\}$ for backpropagation. We sample u_a from the categorical policy in the forward pass and use x_a in the backward pass, hence $\nabla_{\phi_a} u_a \approx \nabla_{\phi_a} x_a$. The resulting gradient estimator is

$$\nabla_{\phi_a} J(\phi_a) \approx \mathbb{E}_B [\nabla_{\phi_a} \mathbf{x} \nabla_{\mathbf{x}} Q_{tot}^\mu(s, \tau, x_1, \dots, x_n)]. \quad (30)$$

c) UT2-P learns a best-individual coordination envelope from replay using the same self-imitation principle as UT2-Q. Returns



Fig. 6. The environments used for evaluation.

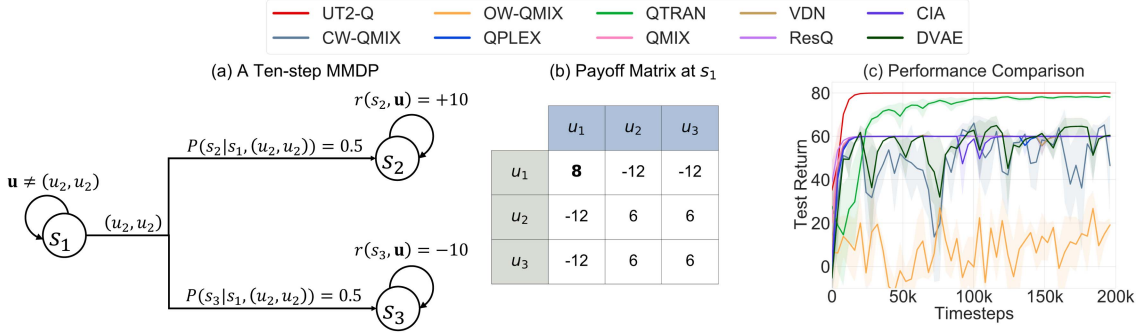


Fig. 7. Performance comparison with value-based baselines on a ten-step Multi-agent Markov Decision Process.

for envelope learning are constructed from predicted mean rewards so that the improvement weights reflect expectation-level coordination quality rather than single noisy realizations. The learned envelope is then used to selectively project critic targets during centralized training. Targets that are more susceptible to optimistic inflation are conservatively replaced by a monotonic surrogate induced by the best-individual envelope. This preserves reliable supervision for the critic that drives policy gradients and extends Target Transformation to continuous and large action spaces without requiring explicit joint-action maximization. Target networks with soft updates are used for stability, consistent with standard actor-critic practice.

The underlying TD target used inside the projection is computed with standard target networks

$$Q_{tot}^{\mu}(s, \mathbf{u}) = r(s, \mathbf{u}) + \gamma \max_{\mathbf{u}'} Q_{tot}^{\mu}(s', \tau', \mathbf{u}'; \phi'_g, \phi'_q),$$

and the target-network parameters ϕ'_g and ϕ'_q are updated via soft updates for stability.

VI. RELATED WORK

We review value decomposition methods for cooperative MARL under centralized training with decentralized execution. A central challenge is credit assignment. A shared team reward must be attributed to individual agents in a way that supports stable learning. Value decomposition addresses this by learning a joint action-value function and factorizing it into agent-wise utilities through a mixing function, while preserving consistency between decentralized greedy execution and centralized greedy selection. This consistency is commonly formalized by the Individual-Global Max principle [14].

VDN [35] and QMIX [15] are representative instantiations that enforce additivity or monotonicity. Many extensions refine the mixer or the utilities without changing the monotonic constraint, including attention-based mixing in Qatten [40], attribution-based decomposition in QPD [41], multi-step targets in SMIX(λ) [42], role-based factorization in RODE [26], deconfounding in DVD [43], and contrastive credit assignment in CIA [44]. These methods improve optimization and generalization, yet the monotonic constraint still restricts the representable class of joint action-values and can lead to suboptimal greedy behavior when optimal targets are non-monotonic [14], [16], [17].

A complementary line of work aims to enlarge the representable class of Q_{tot} by explicitly modeling interactions beyond monotonic mixing. QPLEX [28] introduces a dueling-style decomposition with action-dependent terms to better capture non-monotonicity, and Tesseract [29] models higher-order interactions using low-rank tensor approximations. These approaches improve expressiveness but typically increase centralized complexity as the number of agents and interactions grows, which can make learning and evaluation more difficult in large-scale settings.

Another approach factors centralized evaluation through structured agent interaction models. Deep coordination graph methods [30] and subsequent variants such as DICG [45], CASEC [33], and SOP-CG [46] learn sparse or adaptive interaction structures and perform message passing to approximate joint action selection. These methods provide richer interaction modeling than purely monotonic mixing, but practical deployment can face additional communication overhead. Loopy message passing can also be sensitive to approximation error and may not always converge reliably.

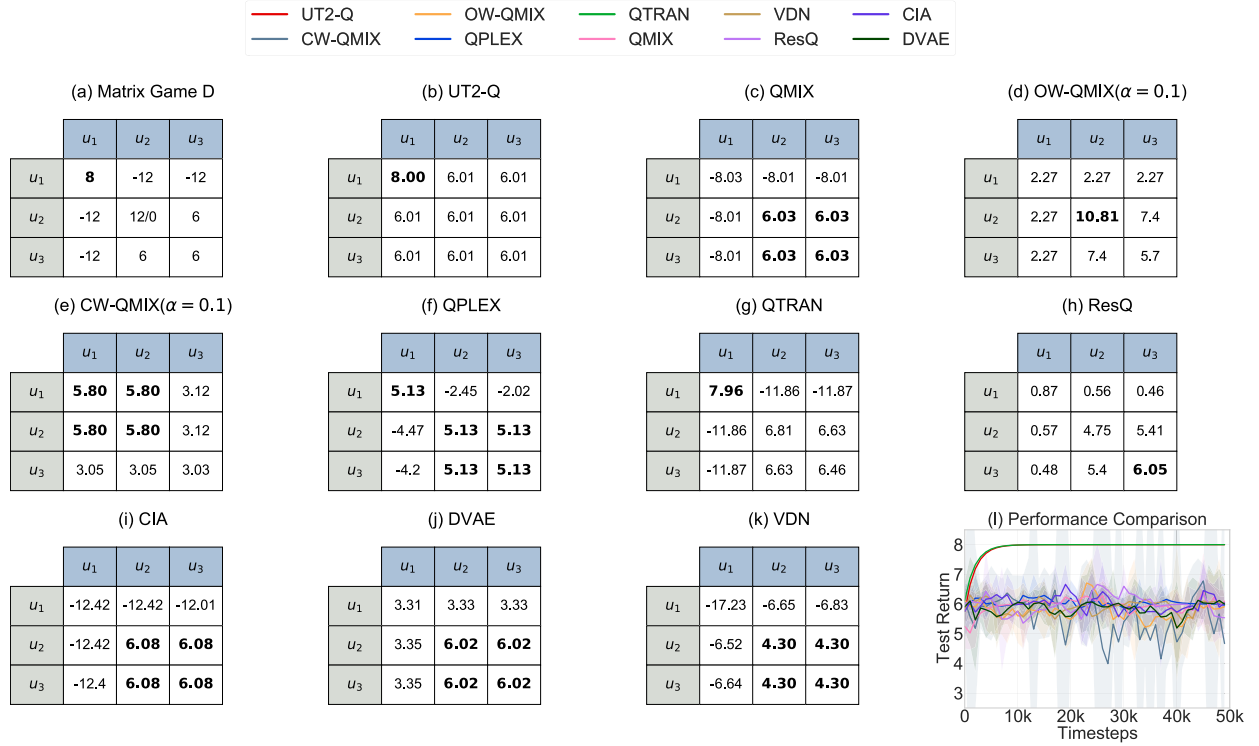


Fig. 8. Performance comparison with value-based baselines on Matrix Game D. The entry for (u_2, u_2) yields 12 with probability 0.5 and 0 otherwise. Boldface highlights the optimal joint action and the greedy joint action u^g induced by each method.

A different family of methods attempts to mitigate representational mismatch by biasing learning toward high-value joint actions. QTRAN [14] uses two soft regularisations to align the greedy action selections between the joint Q -value and the individual values. WQMIX [17] introduces a weighting mechanism to place more importance on better joint actions, similar to Distributed Q -learning [47] and Hysteretic Q -learning [48]. LMRL2 [49] demonstrates that the weighting function is susceptible to misleading stochasticity and utilizes leniency to allow agents to forgive miscoordination during initial exploration but become less lenient with increased state-action pair visitation. QTRAN can be seen as a variant of WQMIX, using a weight of 0 for joint Q -values whose targets are smaller than the current estimate and the chosen action is not greedy. QTRAN approximates $\hat{Q}^*$ as the target instead of the original target Q and thus can deal with stochasticity. However, QTRAN struggles to scale to more complex tasks due to the 0 weight for overestimated Q -values [31]. ResQ [50] masks some state-action value pairs from a joint state-action value function, transforming it into the sum of a main function and a residual function. ResQ can be viewed as an extension of QTRAN, with the residual term having more expressive capacity due to additional joint action information. DAVE [51] uses an ego policy for action selection and an alter ego value function, breaking the IGM principle through an explicit search procedure. GVR [52] proposes inferior target reshaping and superior experience replay to eliminate non-optimal self-transition nodes. While these strategies can improve performance in non-monotonic tasks, optimistic weighting or selective emphasis can still amplify lucky yet suboptimal

returns in stochastic environments, which risks overestimation and unstable convergence.

UT2 addresses non-monotonicity and stochasticity from a different angle. Instead of increasing mixer expressiveness or relying on optimistic reweighting, UT2 constructs an uncertainty-aware target projection that maps non-monotonic and stochastic learning targets into a monotonic-representable surrogate while preserving the optimal joint action. This preserves the CTDE deployment advantages of monotonic value decomposition and mitigates overestimation induced by lucky but suboptimal stochastic realizations during target shaping.

VII. EXPERIMENTS

This section evaluates UT2 to answer the following questions: (1) Can UT2 recover the optimal joint policy when the target joint action value is non-monotonic and stochastic? (2) Is UT2 superior to the existing MARL methods in complex cooperative scenarios? (3) Is explicit target projection necessary compared with methods that only bias learning toward high-value actions through reweighting? (4) How do the key components of UT2 contribute, and what happens when the unique-optimum assumption is violated.

A. Experiment Settings

We evaluate UT2 on six cooperative benchmarks, including Matrix Games, Predator Prey (PP) [30] and Predator Stag Hare (PSH), SMAC [31], SMACv2 [32], and Multi-Agent Coordination Challenge (MACO) [33]. These benchmarks span different

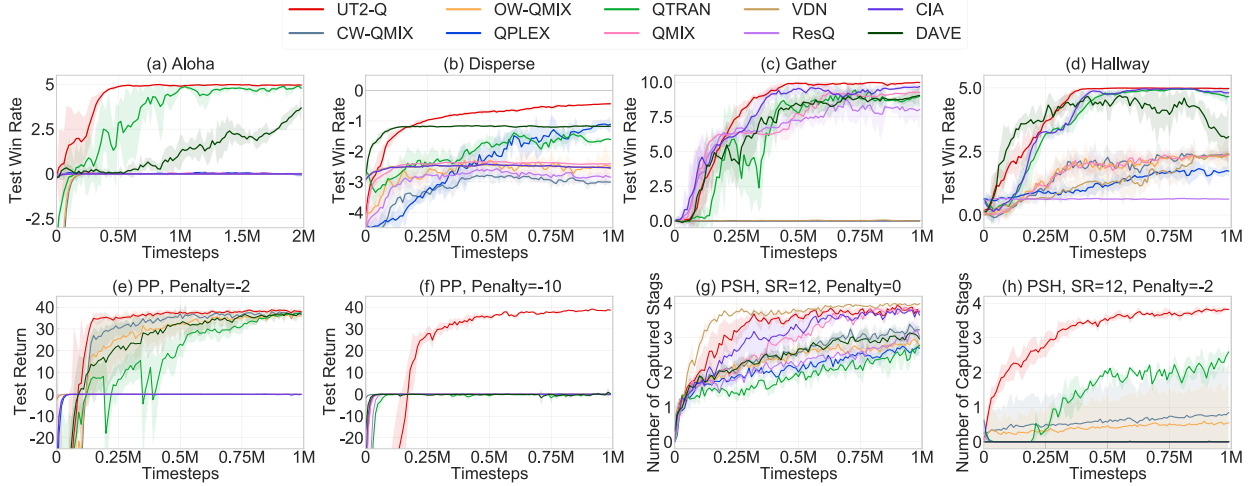


Fig. 9. Performance comparison with value-based baselines on PP, PSH and MACO benchmarks.

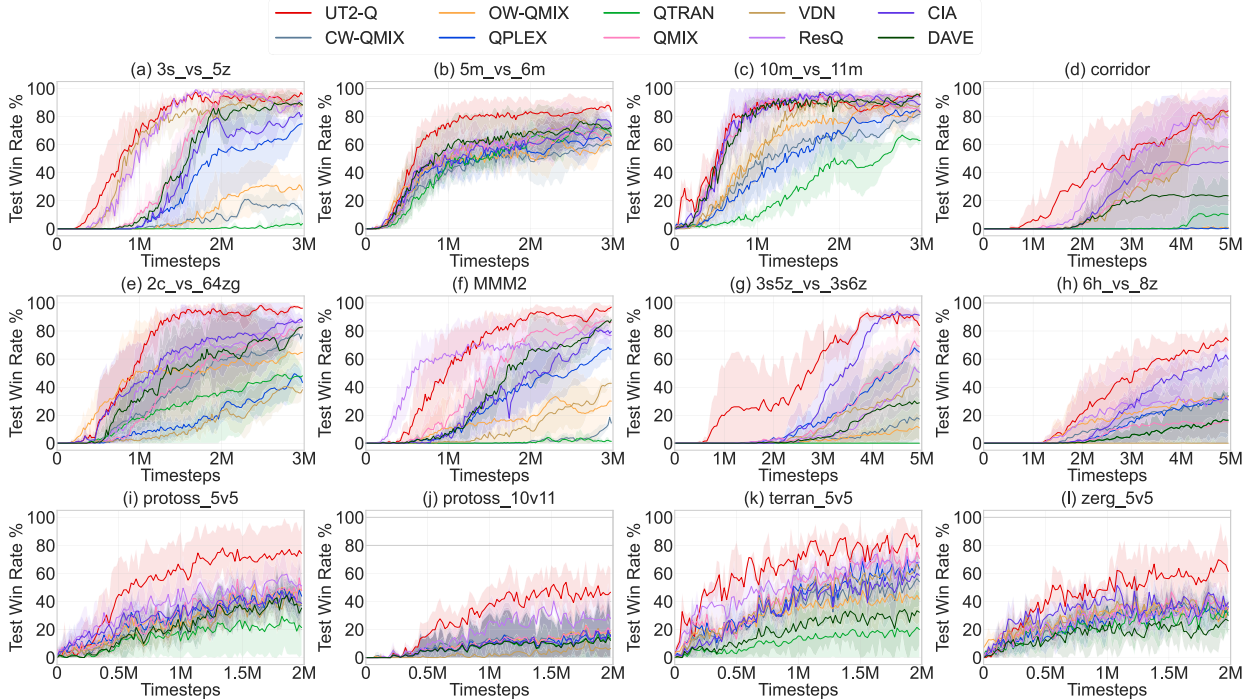


Fig. 10. Performance comparison with value-based baselines in different scenarios on SMAC and SMACv2 benchmarks.

levels of partial observability, coordination difficulty, reward stochasticity, and action-space characteristics, providing a comprehensive assessment of effectiveness, stability, and generality. PP and PSH are canonical benchmarks for relative overgeneralization, which is also emphasized by MACO. Together they provide complementary validation of the same learning pathology.

For value decomposition baselines, we compare UT2-Q with VDN [35], QMIX [15], QTRAN [14], WQMIX [17], QPLEX [28], ResQ [50], CIA [44], and DAVE [51]. For policy-based baselines, we compare UT2-P with MADDPG [10], MASAC [53], COMA [11], DOP [54], MAPPO [55], and MAT [56]. Unless stated otherwise, results report the mean

with a 95% confidence interval over five independent runs with different random seeds. We use official implementations for baseline methods whenever they are available.

B. Matrix Games

We begin with Matrix Game D in Fig. 8, where the optimal joint action is separated from suboptimal alternatives by substantial miscoordination penalties, and the optimal payoff is stochastic. To factor out exploration and sample-complexity effects, we use uniform exploration to approximate a near-uniform dataset over joint actions. We evaluate the stability and convergence of learned policies by test return and its variability.

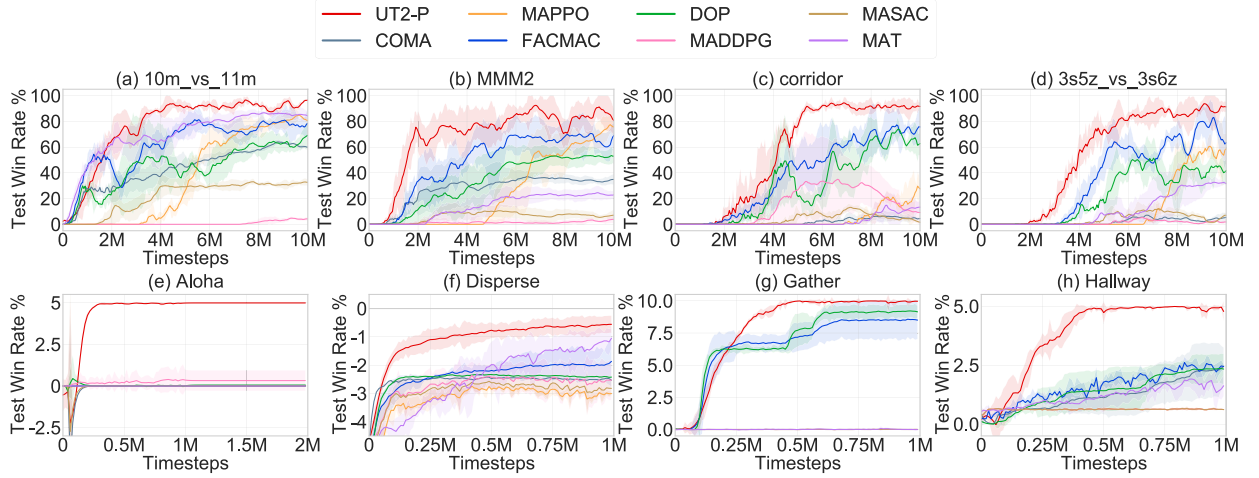


Fig. 11. Performance comparison with policy-based baselines in different scenarios on SMAC and MACO benchmarks.

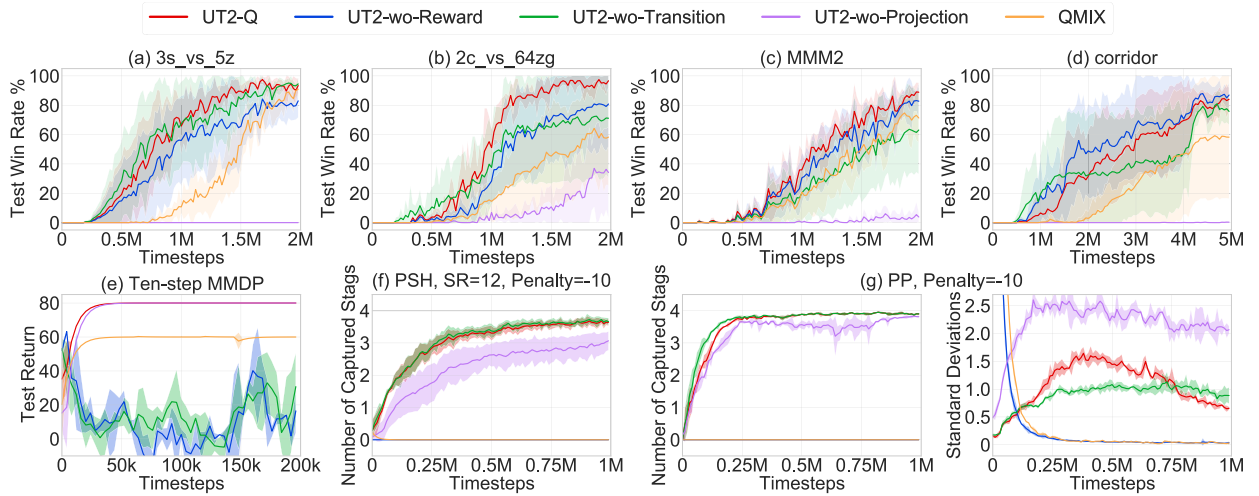


Fig. 12. Ablation studies of key components of UT2-Q.

Fig. 8 shows that UT2-Q consistently recovers the optimal joint action $\mathbf{u}^*$ with low variance. QTRAN also reaches $\mathbf{u}^*$ in this small game, while weight-based baselines are less stable under stochastic targets and often converge to suboptimal coordination patterns. This behavior is consistent with the analysis in Section III-B, where optimistic reweighting can amplify lucky but suboptimal realizations.

We further consider a ten-step MMDP with non-monotonicity and stochastic transitions, shown in Fig. 7. Two agents start at s_1 . Executing $\mathbf{u} = (u_2, u_2)$ at s_1 transitions to s_2 with probability 0.5 and to s_3 otherwise, while other joint actions keep the agents at s_1 . Rewards at s_2 and s_3 are deterministic, with +10 and -10 respectively for any joint action. This construction yields targets that are non-monotonic at s_1 and stochastic through transition randomness. As shown in Fig. 7, methods that rely on monotonic factorization struggle to represent the non-monotonic target ordering at s_1 , while reweighting-based methods remain sensitive to stochastic lucky outcomes and may overemphasize suboptimal actions. UT2-Q achieves optimal performance and

converges faster than QTRAN, supporting the necessity of explicit target projection in stochastic non-monotonic settings.

C. Maco, Pp, and Psh

We next evaluate UT2-Q in PP, PSH, and MACO. These benchmarks combine partial observability with coordination-induced learning pathologies such as relative overgeneralization.

MACO: MACO provides a structured suite of partially observable coordination tasks designed to stress relative overgeneralization under diverse task templates. We train UT2-Q and all value-based baselines on MACO for three million environment steps under the same evaluation schedule and random-seed protocol used for other value-based benchmarks. Fig. 9(a)-(d) shows that UT2-Q improves both performance and stability over strong value-decomposition baselines across MACO tasks, supporting that target projection remains effective in the regime emphasized by MACO.

PP: PP is a partially observable grid world with eight predators and eight prey on a 10×10 grid. Each agent observes a local 5×5 region and can move or attempt to catch. The team receives a reward when two agents coordinate to catch a prey, and incurs a miscoordination penalty when only one agent attempts to catch after surrounding. Fig. 9(e) shows that when the penalty is mild, several baselines can learn positive returns. As the penalty increases, performance becomes substantially more sensitive to representational limitation and training instability. UT2-Q remains reliable in the high-penalty regime, indicating that target projection helps maintain stable credit assignment under non-monotonic coordination incentives.

PSH: PSH introduces stochasticity in both rewards and transitions by adding hares with probabilistic capture outcomes and symmetric punishments, while stags follow the deterministic capture rule. Fig. 9(g)–(h) shows that when the miscoordination penalty is non-zero, multiple baselines fail to consistently learn the high-coordination behavior required to capture stags. UT2-Q learns quickly and reliably across these settings, consistent with its design goal of suppressing lucky but suboptimal target inflation while preserving reliable learning signals through selective target projection.

D. StarCraft Multi-Agent Challenge

We evaluate UT2-Q on SMAC and SMACv2, where tasks require sustained coordination under partial observability and, for SMACv2, increased stochasticity and scenario diversity.

Across SMAC scenarios in Fig. 10(a)–(g), UT2-Q achieves higher win rates and faster convergence than the baselines. Methods such as ResQ and DAVE improve over VDN and QMIX in certain cases, but they are less consistent in scenarios that demand tight coordination or exhibit stronger non-stationarity from decentralized exploration. On SMACv2, UT2-Q maintains superior performance, indicating robustness to increased stochasticity and complexity.

E. Performance Comparison for Policy-Based Methods

To assess generality beyond discrete actions and explicit joint-action maximization, we evaluate UT2-P against policy-based baselines on SMAC and MACO.

Fig. 11 reports results on SMAC and MACO. UT2-P consistently improves over policy-based baselines across scenarios, suggesting that target transformation can be integrated into actor-critic training to enhance robustness in complex cooperative environments.

F. Ablation Studies

We ablate key components of UT2-Q to quantify their contributions. We consider four variants. **UT2-wo-projection** removes target projection and the monotonic joint value function module. It trains the same world-model uncertainty estimator and the same best-individual coordination envelope $\tilde{Q}_a(s, u_a)$, and it selects actions using the greedy rule induced by $\tilde{Q}_a(s, u_a)$. **UT2-wo-Transition** removes the transition uncertainty model. **UT2-wo-Reward** removes the reward uncertainty model. QMIX serves as the natural baseline and

trains a monotonic value decomposition by temporal-difference regression without auxiliary modules.

Fig. 12 shows that removing the reward model leads to overestimation of low-probability high-return outcomes, which degrades performance on tasks with stochastic rewards such as the ten-step MMDP and PSH. Removing the transition model can still handle stochastic rewards, but it fails in tasks where stochasticity primarily enters through transitions, such as the MMDP. Removing target projection yields slower and less reliable learning in high-coordination benchmarks, particularly in SMAC, supporting the necessity of explicit target projection beyond learning an improved best-individual estimator alone.

We also study sensitivity to the uncertainty thresholds in (20). Fig. 13 shows that overly small thresholds can cause insufficient target transformation, while overly large thresholds can induce excessive projection and suppress useful learning signals. In practice, the thresholds depend on reward and state scales. Normalizing observations, states, and rewards reduces tuning burden and improves transferability across tasks.

G. Computational Overhead and Capacity-Matched Controls

UT2 introduces additional training components to construct projected targets, including a world-model uncertainty estimator and a best-individual coordination envelope. We implement the world model using a Multi-Agent Transformer with 6 layers, embedding dimension 64, and 8 attention heads, trained for 3 epochs with Adam, learning rate 1×10^{-3} , betas (0.9, 0.95), weight decay 0.1, and gradient-norm clipping 1.0. This increases training time by approximately 4 to 8 GPU-hours compared with the corresponding baselines. The additional cost provides the uncertainty signal and the coordination envelope that are required by target projection, and it is directly used to stabilize learning and preserve optimality in non-monotonic and stochastic regimes.

We further verify that the gains of UT2 are not explained by increasing the parameter budget of monotonic value decomposition baselines. We construct two capacity-matched controls: 1) **Transformer-QMIX** replaces the QMIX utility network backbone with a Transformer-style encoder and uses an additional prediction head to output per-agent utilities, while keeping the monotonic mixing constraint unchanged, 2) **QMIX-1000 M** scales the QMIX backbone and mixer to approximately one billion parameters while preserving the same monotonic value decomposition structure. Fig. 14 shows that both capacity-matched variants still fail to resolve the non-monotonic and stochastic target mismatch emphasized in this paper and exhibit similar suboptimal convergence behavior to QMIX. This supports that UT2 addresses a structural limitation induced by the monotonic constraint rather than a parameter-capacity shortage.

VIII. LIMITATIONS AND ASSUMPTION RELAXATION

This section discusses how relaxing Assumption 1 affects both the theory and practice of Target Transformation. The strongest policy-preservation guarantee in Section IV relies on a unique optimal joint action at each state. When the optimal joint action is not unique, the optimal set can contain multiple coordination modes. Under decentralized greedy execution

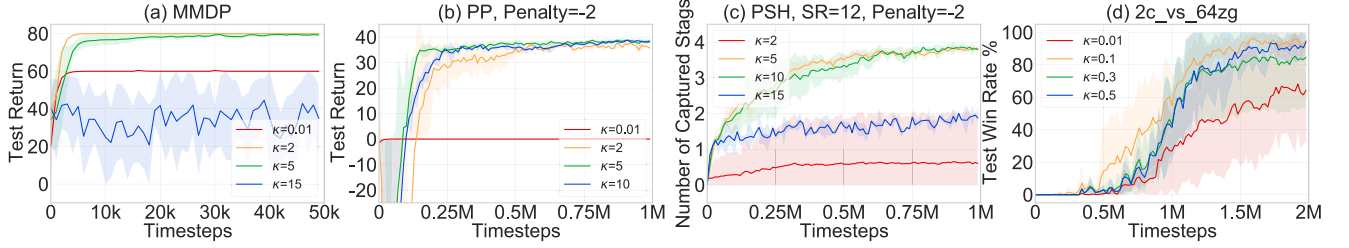


Fig. 13. The results of UT2-Q with different hyperparameter κ .

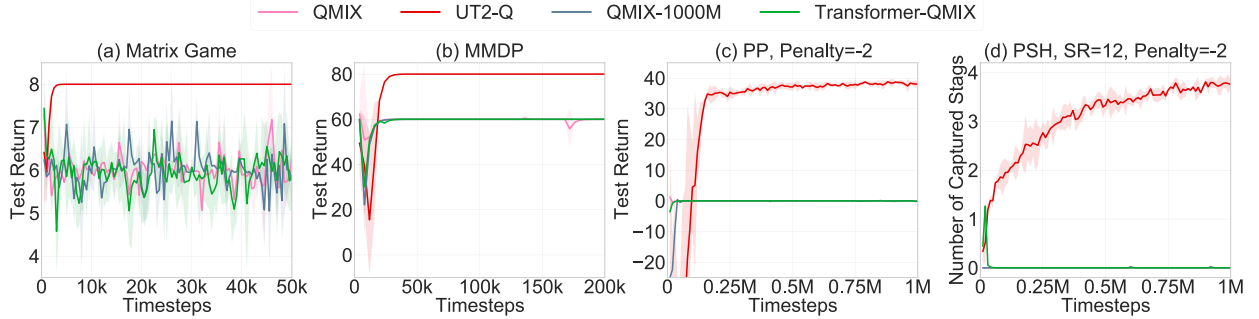


Fig. 14. Capacity-matched controls for monotonic value decomposition.

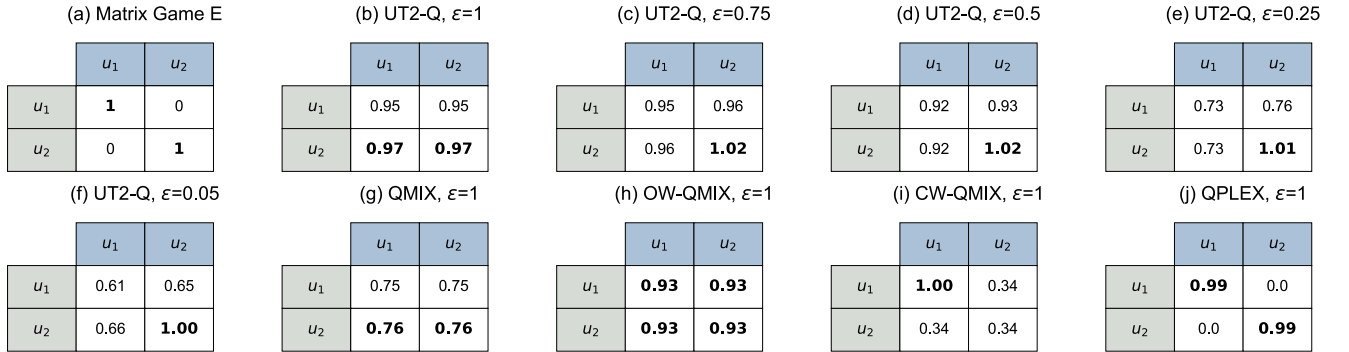


Fig. 15. Performance comparison on Matrix Game E with multiple optimal joint policies.

with the Individual-Global Max principle, instability can arise when the optimal set is not closed under Cartesian products of agent-wise optimal components. This limitation is not specific to UT2. It is inherent to value decomposition methods that rely on decentralized greedy execution under IGM.

A. Multiple Optima and Mode Mixing Under IGM

Fix a state s and define the optimal set $\mathcal{U}^*(s) \triangleq \arg \max_{\mathbf{u} \in \mathcal{U}^n} \mathbb{E}[\mathbf{Q}_{tot}(s, \mathbf{u})]$. Under IGM, decentralized greedy execution induces agent-wise greedy sets $\mathcal{G}_a(s) \triangleq \arg \max_{u_a} Q_a(\tau_a, u_a)$, and the realizable greedy set $\mathcal{G}(s) \triangleq \mathcal{G}_1(s) \times \dots \times \mathcal{G}_n(s)$. A necessary condition for decentralized greedy execution to be always optimal at s is $\mathcal{G}(s) \subseteq \mathcal{U}^*(s)$.

When $|\mathcal{U}^*(s)| > 1$, this condition can fail even if every locally greedy action appears in at least one globally optimal joint action. The core issue is mode mixing. Decentralized greedy execution forms joint actions by Cartesian products. If $\mathcal{U}^*(s)$ is not closed under Cartesian products of agent-wise optimal

components, then mixed joint actions can appear in $\mathcal{G}(s)$ while being strictly suboptimal.

A concrete illustration is a one-step matrix game at state s with two agents and two actions per agent. Let $\mathcal{U}^*(s) = \{(u_1, u_1), (u_2, u_2)\}$, and assume the cross actions (u_1, u_2) and (u_2, u_1) are strictly suboptimal. If an IGM decomposition makes both optimal modes locally greedy, then $u_1 \in \mathcal{G}_a(s)$ and $u_2 \in \mathcal{G}_a(s)$ for each agent. This yields $\mathcal{G}(s)$ that contains (u_1, u_2) and (u_2, u_1) , contradicting $\mathcal{G}(s) \subseteq \mathcal{U}^*(s)$. Therefore, once multiple optimal modes exist and cross-mode combinations are suboptimal, no method that relies on IGM can guarantee that decentralized greedy execution always selects only optimal joint actions at s .

This impossibility matters because IGM is the cornerstone of value decomposition under CTDE. If IGM is abandoned, the maximizer of the centralized joint value function and the maximizers of the per-agent utilities are no longer aligned. Then the optimal action selected by a centralized Q_{tot} is generally different from the joint action formed by decentralized greedy

selection over $\{Q_a\}$. In that regime, decentralized execution is no longer supported by greedy local action selection without additional coordination mechanisms such as explicit joint action search, additional communication, or centralized execution.

We empirically verify the mode-mixing phenomenon in Fig. 15(a), where multiple optimal coordination modes coexist and cross actions are strictly suboptimal. Under a uniform data regime, namely, $\epsilon = 1$, symmetric parameterization and a symmetric squared regression objective admit stationary solutions where each agent assigns equal greedy utility to the actions of different modes. This yields non-singleton $\mathcal{G}_a(s)$ and produces suboptimal mixed joint actions through the Cartesian product $\mathcal{G}(s)$.

A practical mitigation is symmetry breaking in the replay distribution. Under non-uniform data regimes such as decaying ϵ -greedy exploration, the replay distribution concentrates around one coordination mode. This breaks symmetry in the effective learning objective and leads the learned utilities to collapse their greedy sets to singletons, yielding a greedy joint action $\mathbf{u}^g$ that consistently selects one optimal mode. The selected mode is generally run dependent because early trajectories and initialization determine which mode is reinforced by sampling bias.

B. Practical Mitigation by Margin-Based Projection

When Assumption 1 is violated, UT2 can inherit the same IGM limitation because it still trains a monotonic joint value function and executes decentralized greedy actions. In this regime, UT2 does not aim to preserve all optimal modes simultaneously. The practical goal is to promote a consistent mode by suppressing near-tied deterministic alternatives that can sustain mode mixing.

A practical mitigation is to introduce a margin in the deterministic keep condition and to project near-tied deterministic targets more aggressively. Replace (12) by

$$\mathbf{Q}_{tot}^\Lambda(s, \mathbf{u}) = \begin{cases} \mathbf{Q}_{tot}(s, \mathbf{u}), & (s, \mathbf{u}) \in \hat{\mathcal{V}}, \\ \min_a \mathbf{Q}_a(s, u_a), & \text{otherwise,} \end{cases} \quad (31)$$

where $\hat{\mathcal{V}} \triangleq \{(s, \mathbf{u}) \mid \mathbb{I}(s, \mathbf{u}) = 0 \text{ and } \mathbf{Q}_{tot}(s, \mathbf{u}) - Q_{tot}(s, \mathbf{u}) > \delta_Q\}$, and $\delta_Q > 0$ is a small margin. This rule keeps a deterministic target unchanged only when it is clearly underestimated. Otherwise it is projected to $\min_a \mathbf{Q}_a$, which remains compatible with monotonic mixing and tends to suppress competing near-tied modes in the transformed space.

By requiring a strict underestimation margin, UT2 keeps only those deterministic targets that provide a strong corrective signal and projects the remaining near-tied deterministic targets to the conservative envelope surrogate. This shrinks the set of unprojected near-optimal modes in the transformed target space, encourages the learned utilities to collapse onto a single consistent coordination mode, and therefore reduces the chance that decentralized greedy execution mixes local components from different modes.

The main trade-off is that δ_Q is scale dependent. If the reward scale is small or the value gaps among optimal modes are below δ_Q , the margin can suppress a true optimal mode and slow learning or bias mode selection. In practice, δ_Q should be chosen

relative to the typical TD error magnitude. A robust heuristic is to set δ_Q proportional to a moving average of $|\mathbf{Q}_{tot} - Q_{tot}|$ so that the margin adapts to task scale. Reward and observation normalization further reduces sensitivity and makes the margin more transferable across tasks.

IX. CONCLUSION

This paper introduces Target Transformation, a principled approach to mitigating the representational limitation of monotonic value decomposition by reshaping learning targets while preserving the optimal joint action under standard assumptions. We identify a class of learning targets that are simultaneously non-monotonic and stochastic, where weighting-based objectives such as WQMIX face an inherent tension and may fail to recover the optimal joint action for any fixed weight choice. Target Transformation resolves this issue by projecting targets into a monotonic-representable space while maintaining optimal-policy invariance under the uniqueness condition.

Building on this idea, we develop Uncertainty-aware Target Transformation with two practical instantiations. Value-based UT2 targets discrete action spaces under monotonic value decomposition. Policy-based UT2 integrates the same target transformation principle into an actor-critic framework to handle continuous or large action spaces. Experiments on Matrix Games, Predator Prey, Predator Stag Hare, StarCraft Multi-Agent Challenge, SMACv2, and Multi-Agent Coordination Challenge show that UT2 improves both performance and stability over representative baselines, with larger gains in more non-monotonic or more stochastic regimes. We further analyze the limitation when multiple optimal joint actions exist and provide practical mitigation strategies. For future work, we plan to explore additional possibilities within the Target Transformation framework and validate the effectiveness of UT2 algorithms on real-world multi-robot cooperation tasks.

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